Shreya Ann Sebastian

Data Science and Machine Learning | MSc Computer Science, TU Delft

Eindhoven, NL +31 6 47495282 shreyaseb@gmail.com
linkedin.com/in/shreya-sebastian github.com/Shreya-Sebastian
Dutch citizen, no sponsorship needed | Available immediately

Profile

MSc Computer Science graduate (TU Delft, Artificial Intelligence track) with a BSc in Computer Science (Leiden, bioinformatics track), working on data science and machine learning. Thesis work in unsupervised learning and statistical modelling on multimodal sensor data, and in 3D deep learning for image classification. Built and deployed Tech Jobs Radar, a public job board now live at techjobsradar.nl. Other work covers large language models, computer vision and natural language processing.

Education

MSc Computer Science 2023 - 2026
Delft University of Technology Artificial Intelligence Technology Track
Thesis: Automated Discovery of Latent Activity Contexts from Multimodal Sensing for Peer-Interaction Analysis in Preschoolers with Hearing Loss.
Supervision: Socially Perceptive Computing Lab, TU Delft, with external supervision from the University of Miami.
Built an unsupervised pipeline fusing UWB spatial tracking with LENA audio. A Gaussian Mixture Model over acoustic and spatial features recovered six interpretable activity contexts, and Dominant Sets clustering on proximity and body orientation detected spatial groups. Per-context linear mixed models comparing children with and without hearing loss revealed inclusion effects that did not appear in the pooled analysis.

BSc Computer Science 2020 - 2023
Leiden University Bioinformatics Track
Thesis: Enhancing Pollen Classification Accuracy through the Use of Whole Stack Images in 3D Convolutional Neural Networks.
Classified *Urticaceae* pollen (*Parietaria* and two *Urtica* classes) from whole microscope Z-stacks instead of 2D projections, with a custom ConvNet3D and a ResNet3D in PyTorch on 600 variable-size stacks. After tuning augmentation and hyperparameters and comparing whole-stack against sub-stack training, the best model, a ResNet3D trained from scratch on full stacks, reached 95% test accuracy.

International Baccalaureate Diploma
International School Eindhoven
HL: Biology, Chemistry, Dutch B | SL: Environmental Systems & Societies, English A (Language & Literature), Mathematics

Projects

Tech Jobs Radar techjobsradar.nl
Personal project, deployed
Live job board that classifies tech roles and extracts structured fields from Dutch career sites, on Kubernetes on AWS.
Tools: Python, FastAPI, Postgres, Kubernetes, AWS

RetroSTC-Net: Retroactive Consolidation for Neural Networks Repo
Personal research project, open source (MIT)
Retroactive weight consolidation inspired by synaptic tagging and capture, evaluated on Split-MNIST against EWC and SI.
Tools: Python, PyTorch

LLM-Panel Annotation Framework Repo
Group research project, TU Delft
Three-LLM panel that annotates meeting transcripts against a codebook and measures inter-model agreement.
Tools: Python, OpenAI, Gemini and Groq APIs

Online Retail Purchase Prediction Repo

Personal project

Databricks pipeline on real retail transactions that segments customers and predicts repeat and category purchases, tested on a later month. Boosted trees raised a category's average precision from 0.51 to 0.70.

Tools: PySpark, Databricks (Delta Lake, Unity Catalog, MLflow, Jobs), scikit-learn

Deep Learning Reproduction Project Repo

Group project, Deep Learning course, TU Delft

Reproduced a multi-view underwater image enhancement paper and wrote the blog post on what replicated.

Tools: Python, PyTorch

Fault-Tolerant Sensor Data Ingestion (Smart Home PoC) Repo

Software Architecture course, TU Delft

Sensor data ingestion validated under simulated network failures and high load, with CI that tests it in Docker.

Tools: Python, RabbitMQ, Pydantic, Docker, GitLab CI

Personal Finance Engine Repo

Personal project

Matches bank statements to merchants and finds subscriptions, with a local tool-calling LLM agent for hidden merchants.

Tools: Python, DuckDB, RapidFuzz, Ollama

Coursework

MSc (TU Delft): Deep Learning, Machine Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Pattern Recognition, Socio-Cognitive Engineering, Social Signal Processing

BSc (Leiden): Algorithms & Data Structures, Probability & Statistics, Bioinformatics, Introduction to Artificial Intelligence, Data Mining, Human-Computer Interaction (HCI)

Skills

Programming: Python, R, MATLAB, SQL, JavaScript, Java, C++, object-oriented programming

Data Science: Data analysis, statistical analysis, hypothesis testing, regression and linear mixed-effects models, feature engineering, clustering, experiment design, error analysis; pandas, NumPy, scikit-learn

Machine Learning: Supervised and unsupervised learning, classification, deep learning (CNNs), Gaussian mixture models, continual learning, model evaluation (precision, recall, F1), ablation studies, explainability; PyTorch, TensorFlow, OpenCV, GPU training with CUDA

LLMs & NLP: Prompt engineering, few-shot prompting, structured JSON outputs, information extraction, tool-calling agents, LLM evaluation, inter-annotator agreement; OpenAI, Gemini and Groq APIs

Data Engineering: ETL and data pipelines, PySpark, Databricks (Delta Lake, Unity Catalog, Jobs), SQLAlchemy, PostgreSQL, SQLite, DuckDB, Redis queues, RabbitMQ, web crawling, deduplication, fuzzy text matching

Software & MLOps: REST APIs (FastAPI), Pydantic, MLflow (experiment tracking, model registry), Docker, Kubernetes (k3s, Helm), Terraform, AWS (EC2, S3, IAM), CI pipelines (GitHub Actions, GitLab CI), evaluation and regression testing in CI, unit testing (pytest), Prometheus, Cloudflare, Git

Collaboration: Agile, Scrum, team projects, technical writing, presenting results

Extracurriculars & Languages

TU Delft: DSBV Punch basketball | **International School Eindhoven:** Varsity Basketball, Student Council, Science Club
English: native | **Dutch:** conversational